Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data analyst and business intelligence engineer with **8+ years delivering production data pipelines, executive dashboards, and analytics systems** at scale - SQL optimization across 200+ queries (30% performance gain), ETL pipelines on **1.6M financial records**, and automated reporting eliminating 80% of manual operational workload. Deep experience in **data governance, data quality controls, and regulatory compliance** across HUD, LIHTC, and federal housing datasets. Built **AI agent systems** with data integration across enterprise applications - LeaseI (8-agent AI platform, 93.6% precision) and AI on FHIR (NLP-driven clinical data retrieval, ~60% reduction in manual retrieval time). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

SQL & Data Transformation | Business Intelligence & Dashboards | Tableau / Power BI / Data Visualization | Data Governance & Data Quality | Executive Reporting | ETL Pipelines & Data Wrangling | AI Agents & Analytics Automation | Regulatory Compliance Analytics

WORK EXPERIENCE

Entrata - PropTech / Real Estate SaaS (100K+ MAU)

Sep 2018 - Aug 2024

Senior Software Engineer - Data Analytics & Business Intelligence

- **Automated 80% of manual reporting workload** via Python and PL/SQL pipelines - end-to-end data extraction, transformation, and delivery across 20+ production modules; defined KPI definitions and data standards for cross-functional stakeholder reporting across compliance, finance, and operations.
- **Optimized 200+ SQL queries** (30% performance gain across production reporting stack); root-cause analysis on complex multi-schema PostgreSQL, MySQL, and Oracle databases; 99.5% uptime over 6+ consecutive months.
- **Built and owned ETL pipelines on 1.6M financial records** - HUD Tax Credit, LIHTC, AMI-based pricing, HCV income limits - with output validation gates, data quality controls, and regulatory compliance guardrails; 100% PHI compliance on AWS S3 migration, zero security incidents.
- **Data governance and quality assurance** across 20+ production modules: enforced data standards, compliance controls, and audit trails for federal and state regulatory reporting; zero compliance violations over 7 years.
- Led microservices redesign (Docker) - 50% scalability improvement; Salesforce integrations, reporting automations, and CRM data sync via API connections; mentored 5+ junior engineers across 300+ peer code reviews.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate - Data Analytics

- Built **multi-source prediction models across 190+ countries** (Python, R) on WHO and UK Road Safety datasets - R^2=0.977; applied Isolation Forest anomaly detection surfacing data quality issues manual review missed across 3 research pipelines.
- Designed **Tableau, ggplot2, and Quarto executive dashboards** for faculty leadership - 40% research comprehension improvement; validated data accuracy and applied data governance standards before every visualization release.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/Data Engineer Intern

- Built an **AI agent system integrating regulatory data sources** - LangGraph pipeline with RAG retrieval over FAA regulatory corpus, confidence-scored outputs, and validation guardrails; delivered across three business verticals.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Delivered technical instruction to 600+ students across Python programming, database design, and SQL querying - weekly lab instruction, one-on-one debugging support, and rubric-aligned grading.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built ETL pipelines for 1.6M HUD/Tax Credit records; data wrangling and transformation across property management modules (PHP, Python, PostgreSQL); cloud migration to AWS S3.

PROJECTS

Leasel: AI Agent Platform - Automated Data Intelligence & Compliance Analytics 2026

Built an **8-agent AI system delivering automated analytics and root-cause investigation** over regulated document corpora: RAG-based data retrieval (91% Precision@1), reproducible evaluation harness (225-query benchmark, LLM-as-judge scoring, regression gating), and confidence-scored outputs with hard data quality boundaries. AI agent handles data extraction and reasoning; deterministic logic enforces every compliance decision. Production precision: 93.6%. Demonstrates **AI-enabled automation integrating enterprise data sources** - the same pattern as AI copilots and automation modules for operational workflows.

Python · LangGraph · FastAPI · RAG · SQLite · LLM-as-Judge · Data Quality Controls · Regulatory Compliance

AI on FHIR: NLP-Driven Medical Records Query System 2024

Built a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical data retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, SQL, PL/SQL, R, JavaScript, TypeScript, PHP, Bash

Business Intelligence & Data Analytics: SQL (PostgreSQL, MySQL, Oracle, Snowflake-adjacent), Data Transformation, Data Wrangling, Data Validation, Data Quality, Data Governance, ETL Pipelines, Automated Reporting, Executive Dashboards, Ad Hoc Analytics, Data Visualization, Root-Cause Analysis, Regulatory Compliance Analytics, Tableau, Power BI, ggplot2, Quarto, Pandas, NumPy, Scikit-learn, XGBoost, Isolation Forest

AI & Automation: AI Agents, Copilots, Analytics Automation, LangGraph, RAG Pipelines, LLM Evaluation, LLM-as-Judge, OpenAI API, FastAPI, ChromaDB, spaCy, BERT

Cloud & Enterprise: AWS (S3, EC2, RDS, Glue), Salesforce (integrations, reporting automations, data sync), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Git, Linux, Agile/Scrum